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2.3 Phospholipid Bilayer

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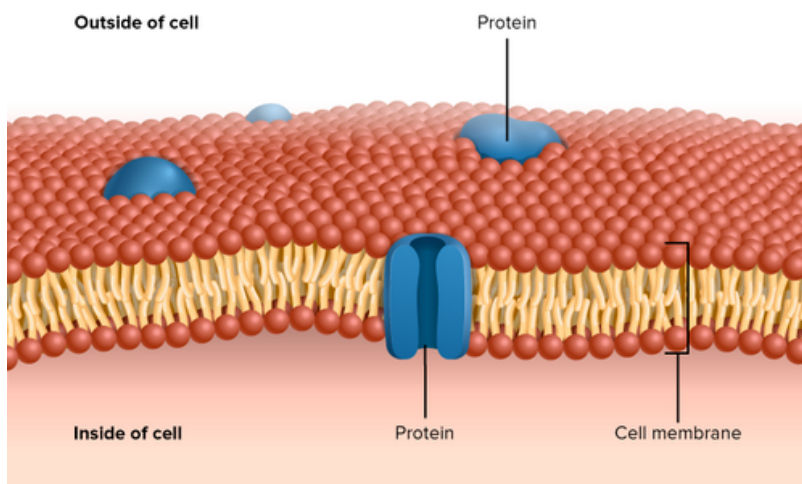
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What you will learn

- Structure of the **cell membrane**
- Role of the phospholipid bilayer in the cell membrane function

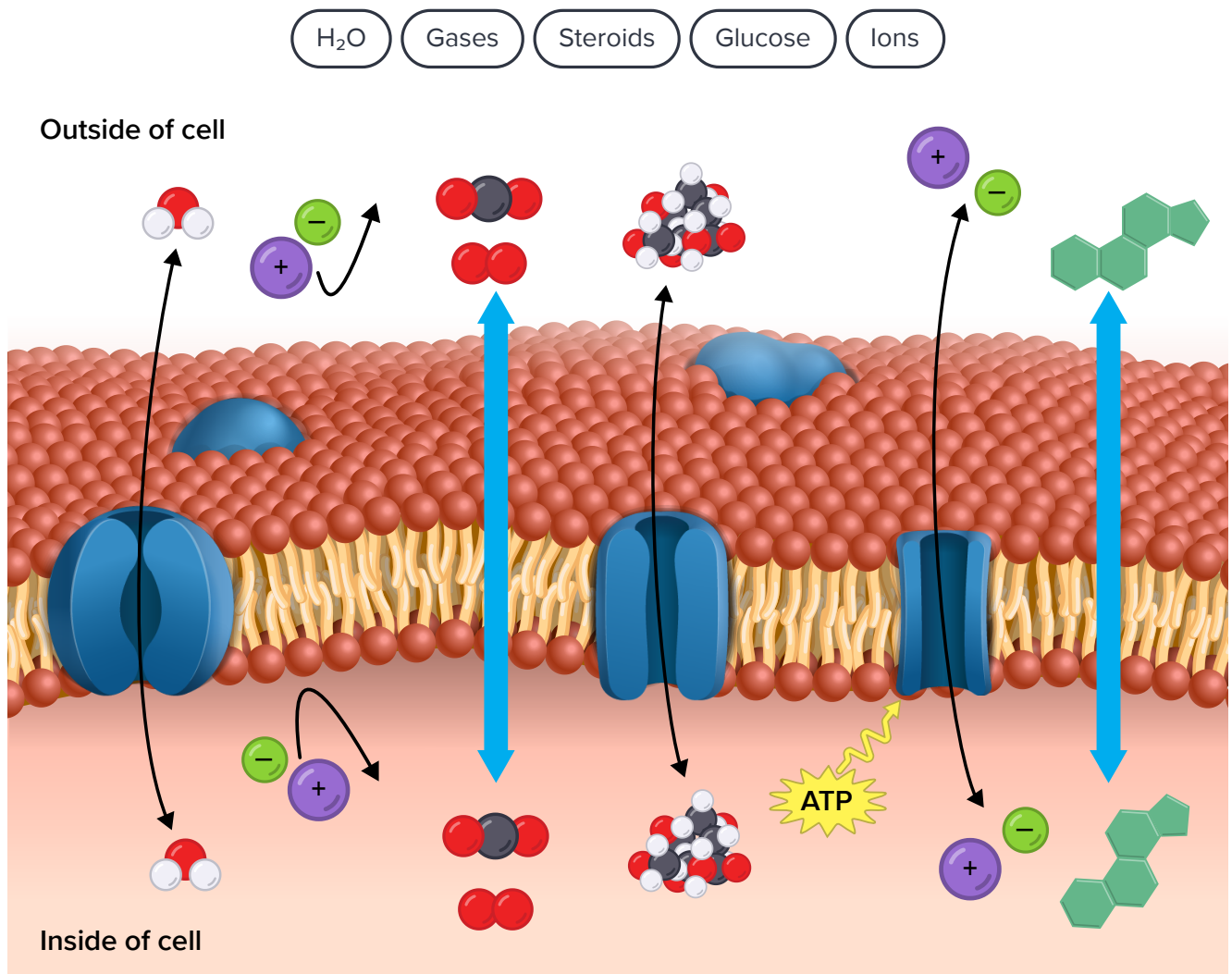


[Figure 1]

All cells have a cell membrane. This membrane surrounds the cell. So what is its role?

Can molecules enter and leave the **cell**? Yes. Can anything or everything enter or leave? No. So, what determines what can go in or out? Is it the **nucleus**? The **DNA**? Or the cell membrane?

that enters and leaves the cell. It allows only certain substances to pass through, while keeping others in or out. The ability to allow only certain molecules in or out of the cell is referred to as selective permeability or **semipermeability**. To understand how the cell membrane controls what crosses into or out of the cell, you need to know its composition.



A Phospholipid Bilayer

The cell membrane is composed mainly of phospholipids, which consist of **fatty acids** and alcohol. The phospholipids in the cell membrane are arranged in two layers, called a **phospholipid bilayer**. Each phospholipid molecule has a head and two tails. The head "loves" **water (hydrophilic)** and the tails "hate" water (**hydrophobic**). The water-hating tails are on the interior of the membrane, whereas the water-loving heads point outwards, toward either the cytoplasm or the fluid that surrounds the cell.

Molecules that are hydrophobic can easily pass through the cell membrane, if they are small enough, because they are water-hating like the interior of the membrane. Small **lipids** and steroids are hydrophobic and can readily cross the membrane. Molecules that are hydrophilic, on the other hand, cannot pass through the cell membrane—at least not without help—because they are water-loving like the exterior of the membrane, and are therefore excluded from the interior of the membrane. Hydrophilic molecules such as **glucose** and **ions** like Na^+ and K^+ need the help of special **proteins** to cross the membrane.



DID YOU KNOW?

In 1925, Gortner and Grendel studied human **red blood cells** to measure the amount of lipids in the cell membrane. Their **observations** suggested that the cell membrane could be a double layer of lipids.

The Cell Membrane

Bozeman Science



Watch on

Science Friday: Candy Corn in Space

Candy corn is a very tasty treat. In this video by **Science Friday**, astronaut Don Pettit uses Candy Corn to demonstrate the effects of hydrophobic and hydrophilic interactions.





Watch on

Summary

- The cell membrane forms a barrier between the cytoplasm and the environment outside the cell. The cell membrane has selective permeability.
 - The cell membrane is primarily composed of phospholipids arranged in a bilayer, with the hydrophobic tails on the interior of the membrane, and the hydrophilic heads pointing outwards.
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Review

1. Describe the role of the cell membrane.
2. What is meant by semipermeability?
3. Describe the composition of the cell membrane.
4. Explain why hydrophobic molecules can easily cross the cell membrane, while hydrophilic molecules cannot.



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